

Paper Practice 2.6 • WORKED KEY**Solving Proportions**

WORKED KEY (odd-numbered items) – Cross-multiply, then solve the equation you get. When the answer isn't a whole number, keep the exact fraction and round only at the end. Even-numbered items are yours to work.

Example 1

Solve each proportion. If necessary, round to the nearest hundredth.

1. $\frac{5}{9} = \frac{g}{27}$

Answer: $27 \div 9 = 3$, so $g = 5 \cdot 3 = \mathbf{15}$ (or cross-multiply: $9g = 135$)

2. $\frac{2}{x} = \frac{4}{28}$

3. $\frac{h}{8} = \frac{16}{30}$

Answer: $30h = 128 \rightarrow h = \frac{128}{30} = \frac{64}{15} \approx \mathbf{4.27}$

4. $\frac{7}{15} = \frac{42}{n}$

5. $\frac{0.4}{w} = \frac{3}{0.75}$

Answer: $3w = 0.4 \cdot 0.75 = 0.3 \rightarrow w = \mathbf{0.1}$

6. $\frac{8.4}{4.8} = \frac{6.3}{k}$

Example 2

Solve each proportion. If necessary, round to the nearest hundredth.

7. $\frac{2b + 14}{8} = \frac{1 - 2b}{12}$

Answer: $12(2b + 14) = 8(1 - 2b) \rightarrow 24b + 168 = 8 - 16b \rightarrow 40b = -160 \rightarrow b = \mathbf{-4}$

8. $\frac{2z - 3}{3} = \frac{3z - 1}{6}$

9. $\frac{9}{-t - 6} = \frac{27}{2t + 22}$

Answer: $9(2t + 22) = 27(-t - 6) \rightarrow 18t + 198 = -27t - 162 \rightarrow 45t = -360 \rightarrow t = \mathbf{-8}$. Check: $9 \div 2 = 4.5$ and $27 \div 6 = 4.5 \checkmark$

10. $\frac{6}{4y + 1} = \frac{9}{9y - 6}$

11. $\frac{5d + 1}{7} = \frac{6d - 5}{15}$

Answer: $15(5d + 1) = 7(6d - 5) \rightarrow 75d + 15 = 42d - 35 \rightarrow 33d = -50 \rightarrow d = \mathbf{-\frac{50}{33} \approx -1.52}$

12. $\frac{4f + 7}{24} = \frac{7f - 3}{32}$

Example 3

13. TRAVEL Carla traveled 320 miles in 5 hours. At this rate, how far will Carla travel in 7 hours?

Answer: $\frac{320 \text{ mi}}{5 \text{ h}} = \frac{x}{7 \text{ h}} \rightarrow 5x = 2240 \rightarrow x = \mathbf{448 \text{ miles}}$ ($64 \text{ mi/h} \times 7 \text{ h}$)

14. GIFT SHOP A gift shop sells souvenir magnets. The store is selling 3 magnets for \$10. At this price, how much will it cost to buy 8 magnets?

15. BASEBALL During the regular season, each Major League Baseball team plays 162 games. In the 2019 regular season, the Cincinnati Reds had a winning percentage of about 46.3%. How many games did the Cincinnati Reds win in the 2019 regular season?

Answer: $\frac{46.3}{100} = \frac{w}{162} \rightarrow 100w = 7500.6 \rightarrow w \approx 75.006 \rightarrow \mathbf{\text{about 75 games}}$